VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Machine Learning (23CS6PCMAL)

Submitted by

Krish Bam (1BM23CS158)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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B.M.S. College of Engineering,

Bull Temple Road, Bangalore 560019

(Affiliated To Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Machine Learning (23CS6PCMAL)" carried out by **Krish Bam (1BM23CS158),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Machine Learning (23CS6PCMAL) work prescribed for the said degree.

Dr.Pallavi G B Assistant Professor Department of CSE, BMSCE	Dr. Kavitha Sooda Professor & HOD Department of CSE, BMSCE

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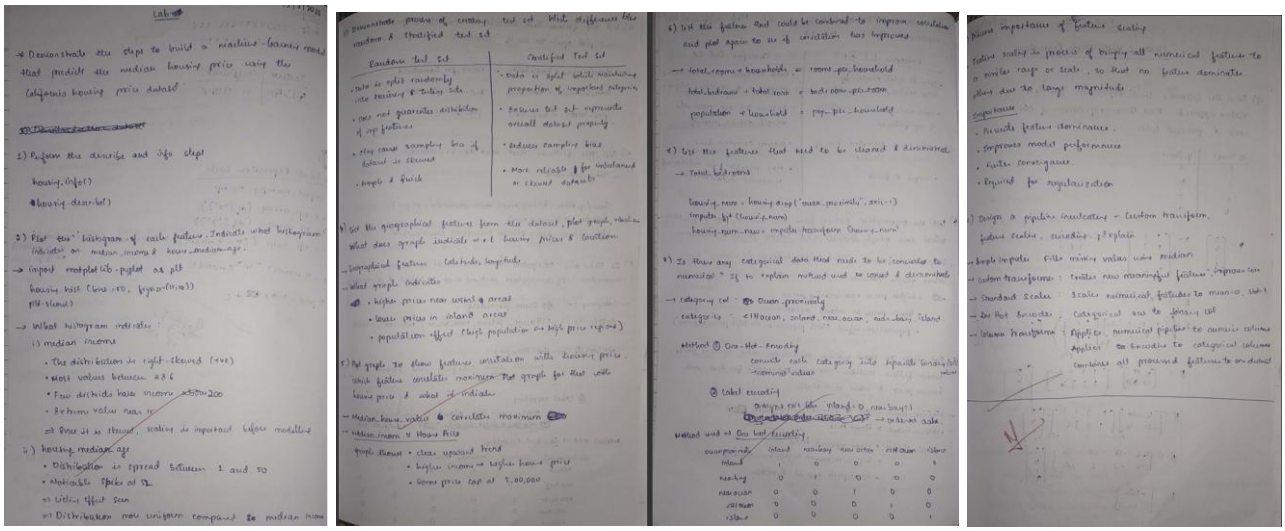
Github Link:
https://github.com/KrishBam/Machine-Learning-Lab.git

Program 1

Write a python program to import and export data using Pandas library functions

Screenshot



Code:

```
import pandas as pd
# Load dataset
housing = pd.read_csv("housing.csv")
# Display first 5 rows
housing.head()
housing.info()
housing.describe()
import matplotlib.pyplot as plt

housing.hist(bins=50, figsize=(15,10))
plt.show()

from sklearn.model_selection import train_test_split

train_set, test_set = train_test_split(housing, test_size=0.2, random_state=42)

import numpy as np

housing["income_cat"] = pd.cut(housing["median_income"],
                               bins=[0., 1.5, 3.0, 4.5, 6., np.inf],
```

```
                        labels=[1,2,3,4,5])

from sklearn.model_selection import StratifiedShuffleSplit

split = StratifiedShuffleSplit(n_splits=1, test_size=0.2, random_state=42)

for train_index, test_index in split.split(housing, housing["income_cat"]):
    strat_train_set = housing.loc[train_index]
    strat_test_set = housing.loc[test_index]

housing.plot(kind="scatter", x="longitude", y="latitude",
             alpha=0.4,
             s=housing["population"]/100,
             label="population",
             c="median_house_value",
             cmap="jet",
             colorbar=True)
plt.legend()
plt.show()

corr_matrix = housing.corr(numeric_only=True)

corr_matrix["median_house_value"].sort_values(ascending=False)

housing.plot(kind="scatter", x="median_income", y="median_house_value", alpha=0.1)
plt.show()

housing["rooms_per_household"] = housing["total_rooms"] / housing["households"]
housing["bedrooms_per_room"] = housing["total_bedrooms"] / housing["total_rooms"]
housing["population_per_household"] = housing["population"] / housing["households"]

corr_matrix = housing.corr(numeric_only=True)
corr_matrix["median_house_value"].sort_values(ascending=False)

housing["total_bedrooms"].isnull().sum()
from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy="median")

housing_num = housing.drop("ocean_proximity", axis=1)

imputer.fit(housing_num)

housing_num_imputed = imputer.transform(housing_num)

housing["ocean_proximity"].value_counts()
from sklearn.preprocessing import OneHotEncoder
```

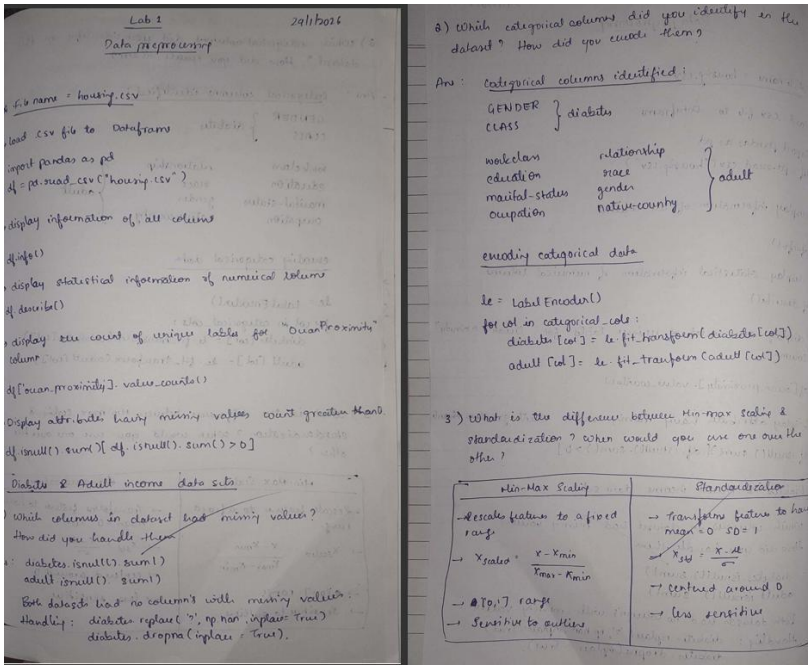
```
housing_cat = housing[["ocean_proximity"]]

encoder = OneHotEncoder()

housing_cat_encoded = encoder.fit_transform(housing_cat)

housing_cat_encoded.toarray()

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

housing_scaled = scaler.fit_transform(housing_num_imputed)

from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.base import BaseEstimator, TransformerMixin

# Custom Transformer
class CombinedAttributesAdder(BaseEstimator, TransformerMixin):
    def __init__(self, add_bedrooms_per_room=True):
        self.add_bedrooms_per_room = add_bedrooms_per_room

    def fit(self, X, y=None):
        return self

    def transform(self, X):
        rooms_per_household = X[:, 3] / X[:, 6]
        population_per_household = X[:, 5] / X[:, 6]

        if self.add_bedrooms_per_room:
            bedrooms_per_room = X[:, 4] / X[:, 3]
            return np.c_[X, rooms_per_household, population_per_household, bedrooms_per_room]
        else:
            return np.c_[X, rooms_per_household, population_per_household]


num_attribs = list(housing_num.columns)
cat_attribs = ["ocean_proximity"]

num_pipeline = Pipeline([
    ('imputer', SimpleImputer(strategy="median")),
    ('attribs_adder', CombinedAttributesAdder()),
    ('std_scaler', StandardScaler())
])

full_pipeline = ColumnTransformer([
    ("num", num_pipeline, num_attribs),
```

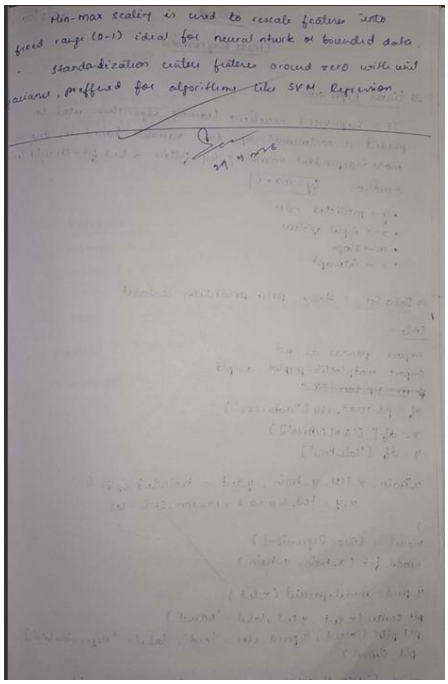
```
    ("cat", OneHotEncoder(), cat_attribs)
])

housing_prepared = full_pipeline.fit_transform(housing)

from sklearn.linear_model import LinearRegression

lin_reg = LinearRegression()
lin_reg.fit(housing_prepared, housing["median_house_value"])
```

Program 2

Demonstrate various data pre-processing techniques for a given dataset

Screenshot





Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OrdinalEncoder, OneHotEncoder
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from scipy import stats
def createdata():
  data = {
      'Age': np.random.randint(18, 70, size=20),
      'Salary': np.random.randint(30000, 120000, size=20),
      'Purchased': np.random.choice([0, 1], size=20),
      'Gender': np.random.choice(['Male', 'Female'], size=20),
      'City': np.random.choice(['New York', 'San Francisco', 'Los Angeles'], size=20)
  }

  df = pd.DataFrame(data)
  return df
df = createdata()
df.head(10)
```

```
df.shape
# Introduce some missing values for demonstration
df.loc[5, 'Age'] = np.nan
df.loc[10, 'Salary'] = np.nan
df.head(10)
# Basic information about the dataset
print(df.info())
# Summary statistics
print(df.describe())
#Code to Find Missing Values
# Check for missing values in each column
missing_values = df.isnull().sum()

# Display columns with missing values
print(missing_values[missing_values > 0])
#Set the values to some value (zero, the mean, the median, etc.).
# Step 1: Create an instance of SimpleImputer with the median strategy for Age and mean stratergy
for Salary
imputer1 = SimpleImputer(strategy="median")
imputer2 = SimpleImputer(strategy="mean")

df_copy=df

# Step 2: Fit the imputer on the "Age" and "Salary"column
# Note: SimpleImputer expects a 2D array, so we reshape the column
imputer1.fit(df_copy[["Age"]])
imputer2.fit(df_copy[["Salary"]])

# Step 3: Transform (fill) the missing values in the "Age" and "Salary"c column
df_copy["Age"] = imputer1.transform(df[["Age"]])
df_copy["Salary"] = imputer2.transform(df[["Salary"]])

# Verify that there are no missing values left
print(df_copy["Age"].isnull().sum())
print(df_copy["Salary"].isnull().sum())
#Handling Categorical Attributes
#Using Ordinal Encoding for gender COlumn and One-Hot Encoding for City Column

# Initialize OrdinalEncoder
ordinal_encoder = OrdinalEncoder(categories=[["Male", "Female"]])
# Fit and transform the data
df_copy["Gender_Encoded"] = ordinal_encoder.fit_transform(df_copy[["Gender"]])

# Initialize OneHotEncoder
onehot_encoder = OneHotEncoder()

# Fit and transform the "City" column
encoded_data = onehot_encoder.fit_transform(df[["City"]])
```

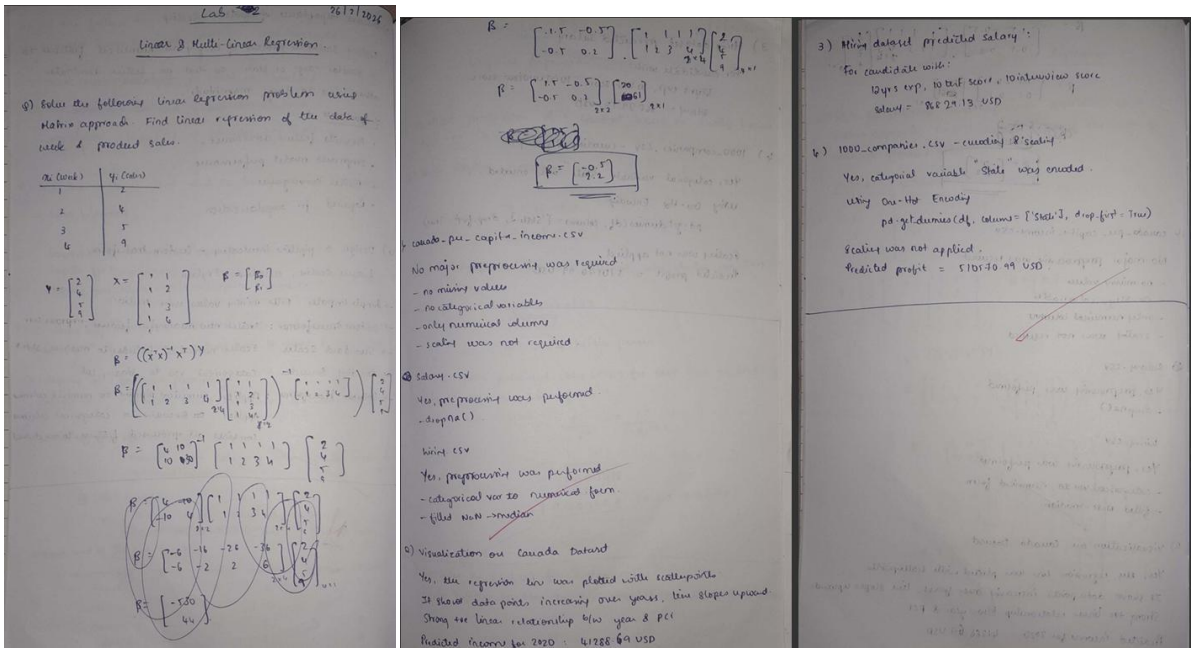
```
# Convert the sparse matrix to a dense array
encoded_array = encoded_data.toarray()

# Convert to DataFrame for better visualization
encoded_df = pd.DataFrame(encoded_array,
columns=onehot_encoder.get_feature_names_out(["City"]))
df_encoded = pd.concat([df_copy, encoded_df], axis=1)

df_encoded.drop("Gender", axis=1, inplace=True)
df_encoded.drop("City", axis=1, inplace=True)

print(df_encoded. head())
#Data Transformation
# Min-Max Scaler/Normalization (range 0-1)
#Pros: Keeps all data between 0 and 1; ideal for distance-based models.
#Cons: Can distort data distribution, especially with extreme outliers.
normalizer = MinMaxScaler()
df_encoded[['Salary']] = normalizer.fit_transform(df_encoded[['Salary']])
df_encoded.head()
# Standardization (mean=0, variance=1)
#Pros: Works well for normally distributed data; suitable for many models.
#Cons: Sensitive to outliers.
scaler = StandardScaler()
df_encoded[['Age']] = scaler.fit_transform(df_encoded[['Age']])
df_encoded.head()

#Removing Outliers
# Outlier Detection and Treatment using IQR
#Pros: Simple and effective for mild outliers.
#Cons: May overly reduce variation if there are many extreme outliers.
df_encoded_copy1=df_encoded
df_encoded_copy2=df_encoded
df_encoded_copy3=df_encoded

Q1 = df_encoded_copy1['Salary'].quantile(0.25)
Q3 = df_encoded_copy1['Salary'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df_encoded_copy1['Salary'] = np.where(df_encoded_copy1['Salary'] > upper_bound, upper_bound,
                np.where(df_encoded_copy1['Salary'] < lower_bound, lower_bound,
df_encoded_copy1['Salary']))

print(df_encoded_copy1.head())


#Removing Outliers
```

```
# Z-score method
#Pros: Good for normally distributed data.
#Cons: Not suitable for non-normal data; may miss outliers in skewed distributions.

df_encoded_copy2['Salary_zscore'] = stats.zscore(df_encoded_copy2['Salary'])
df_encoded_copy2['Salary'] = np.where(df_encoded_copy2['Salary_zscore'].abs() > 3, np.nan,
df_encoded_copy2['Salary'])  # Replace outliers with NaN
print(df_encoded_copy2.head())
#Removing Outliers
# Median replacement for outliers
#Pros: Keeps distribution shape intact, useful when capping isn't feasible.
#Cons: May distort data if outliers represent real phenomena.
df_encoded_copy3['Salary_zscore'] = stats.zscore(df_encoded_copy3['Salary'])
median_salary = df_encoded_copy3['Salary'].median()
df_encoded_copy3['Salary'] = np.where(df_encoded_copy3['Salary_zscore'].abs() > 3,
median_salary, df_encoded_copy3['Salary'])
print(df_encoded_copy3.head())
```

Program 3

Implement Linear and Multi-Linear Regression algorithm using appropriate dataset

Screenshot



Code:

```
# Predict Canada's Per Capita Income for 2020

# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Load dataset
df = pd.read_csv('canada_per_capita_income.csv')

# Display first 5 rows
print(df.head())

# Define independent and dependent variables
X = df[['year']]      # Feature
y = df['per capita income (US$)']

# Create model
model = LinearRegression()

# Train model
```

```
model.fit(X, y)

# Predict income for year 2020
predicted_income = model.predict([[2020]])

print("Predicted Per Capita Income in 2020:", predicted_income[0])

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Year")
plt.ylabel("Per Capita Income")
plt.show()
# ----------------------------------------
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
# ----------------------------------------
# Load dataset
df = pd.read_csv('salary.csv')

# Remove missing values
df.dropna(inplace=True)

# Define variables
X = df[['YearsExperience']]
y = df['Salary']

# Create and train model
model = LinearRegression()
model.fit(X, y)

# Predict salary for 12 years experience
predicted_salary = model.predict([[12]])

print("Predicted Salary for 12 years experience:", predicted_salary[0])

# 🔹 Plot graph
plt.scatter(X, y)             # Actual data points
plt.plot(X, model.predict(X))  # Regression line

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Salary vs Experience (Linear Regression)")

plt.show()
# Import libraries
```

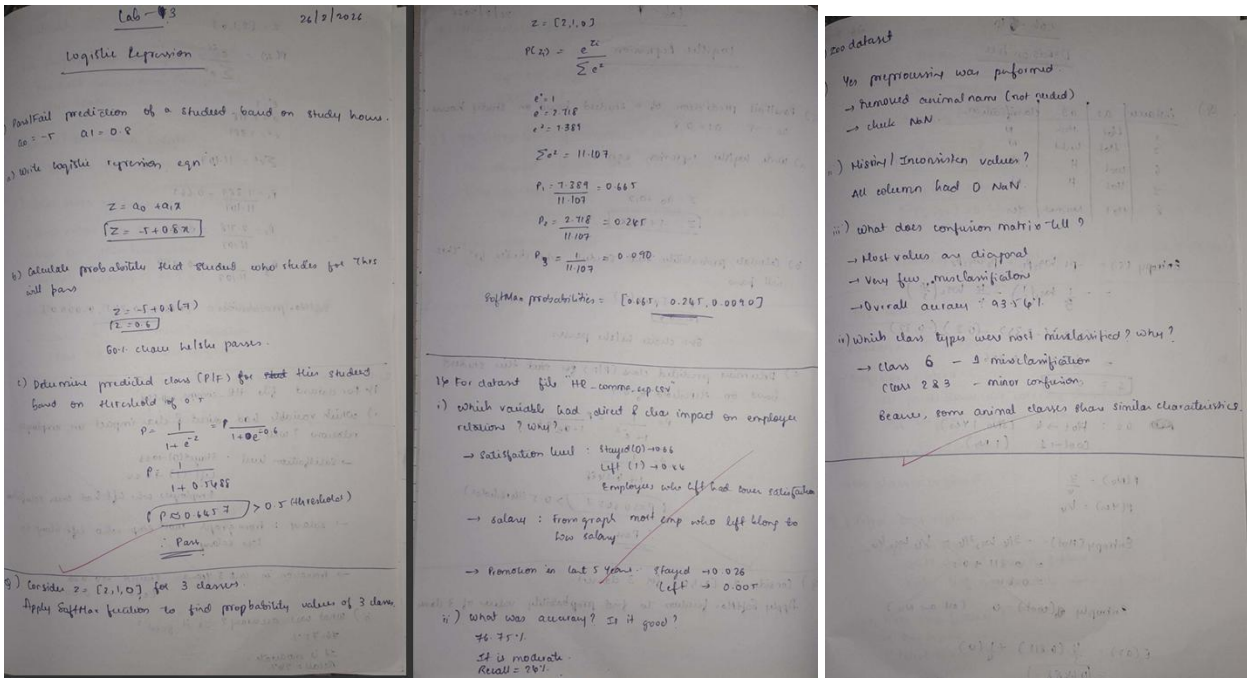
```
import pandas as pd
from sklearn.linear_model import LinearRegression
# ----------------------------------------
# Load dataset
df = pd.read_csv('hiring.csv')

# Convert word numbers to numeric
df['experience'] = df['experience'].replace({
    'zero':0,'one':1,'two':2,'three':3,'four':4,
    'five':5,'six':6,'seven':7,'eight':8,'nine':9,'ten':10
})

# Convert to numeric type
df['experience'] = pd.to_numeric(df['experience'], errors='coerce')

# Fill missing values using median
df['experience'] = df['experience'].fillna(df['experience'].median())
df['test_score(out of 10)'] = df['test_score(out of 10)'].fillna(
    df['test_score(out of 10)'].median()
)

X = df[['experience',
        'test_score(out of 10)',
        'interview_score(out of 10)']]

y = df['salary($)']

model = LinearRegression()
model.fit(X, y)

# Candidate 1
input1 = pd.DataFrame([[2, 9, 6]],
                      columns=X.columns)
salary1 = model.predict(input1)

# Candidate 2
input2 = pd.DataFrame([[12, 10, 10]],
                      columns=X.columns)
salary2 = model.predict(input2)

print("Predicted Salary (2 yr exp, 9 test, 6 interview):", salary1[0])
print("Predicted Salary (12 yr exp, 10 test, 10 interview):", salary2[0])
# 1000 Companies Profit Prediction
# Import libraries
import pandas as pd
from sklearn.linear_model import LinearRegression
# ----------------------------------------
# Load dataset
```

```
df = pd.read_csv('1000_Companies.csv')

# Handle categorical column (State)
df = pd.get_dummies(df, columns=['State'], drop_first=True)

# Define features and target
X = df.drop('Profit', axis=1)
y = df['Profit']

model = LinearRegression()
model.fit(X, y)

# Create input for Florida
# (Assuming columns are: R&D Spend, Administration, Marketing Spend,
# State_Florida, State_New York)

input_profit = pd.DataFrame([[91694.48,
                              515841.3,
                              11931.24,
                              1, 0]],  # Florida = 1
                             columns=X.columns)

predicted_profit = model.predict(input_profit)

print("Predicted Profit:", predicted_profit[0])
```

Program 4

Build Logistic Regression Model for a given dataset

Screenshot



Code:

 # Logiastic regression - binary classification

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# Load Dataset
df = pd.read_csv("HR_comma_sep.csv")

# - - - - - - - - - - - - - - - - - - -
# 1. Exploratory Data Analysis
# - - - - - - - - - - - - - - - - - - -

print("First 5 rows:")
print(df.head())

print("\nDataset Info:")
```

```
print(df.info())

print("\nEmployee Retention Count:")
print(df['left'].value_counts())

print("\nAverage values grouped by retention:")
print(df.groupby('left').mean(numeric_only=True))

# ------------------
# 2. Salary vs Retention (Bar Chart)
# ------------------

pd.crosstab(df.salary, df.left).plot(kind='bar')
plt.title("Salary vs Employee Retention")
plt.xlabel("Salary Level")
plt.ylabel("Number of Employees")
plt.show()

# ------------------
# 3. Department vs Retention (Bar Chart)
# ------------------
pd.crosstab(df.Department, df.left).plot(kind='bar', figsize=(10,5))
plt.title("Department vs Employee Retention")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.xticks(rotation=45)
plt.show()
# ------------------
# 4. Feature Selection
# ------------------
# Selecting important features
df2 = df[['satisfaction_level',
          'average_montly_hours',
          'promotion_last_5years',
          'salary',
          'left']]

# Convert salary (categorical) to dummy variables
df2 = pd.get_dummies(df2, columns=['salary'], drop_first=True)

# ------------------
# 5. Train-Test Split
# ------------------

X = df2.drop('left', axis=1)
y = df2['left']

X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=42)

# ------------------
# 6. Build Logistic Regression Model
# ------------------

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# ------------------
# 7. Prediction & Accuracy
# ------------------

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nModel Accuracy:", accuracy)
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

 # Logiastic regression - multiclass classification
 # Logistic Regression - Multiclass Classification (Zoo Dataset)

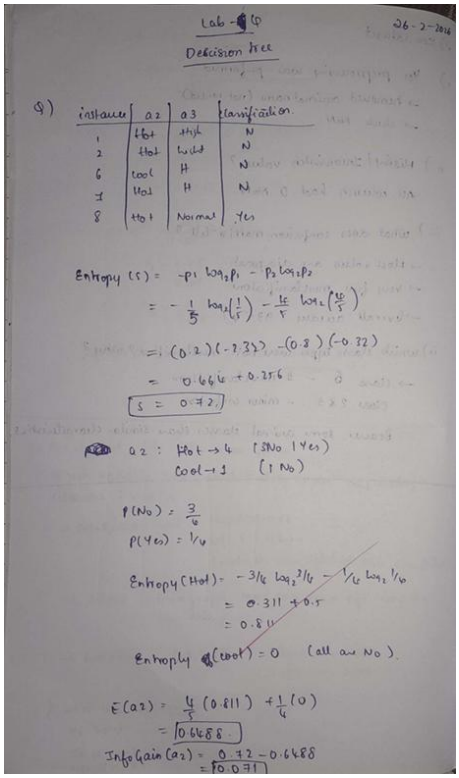
```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

# ------------------
# 1. Load Dataset
# ------------------

zoo = pd.read_csv("zoo-data.csv")
class_type = pd.read_csv("zoo-class-type.csv")

print("First 5 rows:")
print(zoo.head())

# ------------------
# 2. Data Preprocessing
# ------------------

# Drop animal name (not useful for prediction)
zoo = zoo.drop("animal_name", axis=1)
```

```
# Check for null values
print("\nMissing values:")
print(zoo.isnull().sum())

# Features and Target
X = zoo.drop("class_type", axis=1)
y = zoo["class_type"]

# ------------------
# 3. Train-Test Split
# ------------------

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

# ------------------
# 4. Build Logistic Regression Model
# ------------------

model = LogisticRegression(max_iter=1000, multi_class='multinomial')
model.fit(X_train, y_train)

# ------------------
# 5. Prediction
# ------------------

y_pred = model.predict(X_test)

# ------------------
# 6. Accuracy
# ------------------

accuracy = accuracy_score(y_test, y_pred)
print("\nModel Accuracy:", accuracy)

# ------------------
# 7. Confusion Matrix
# ------------------

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix - Zoo Dataset")
plt.show()
```

Program 5

Use an appropriate data set for building the decision tree (ID3) and apply this knowledge to classify a new sample.

Screenshot



Lab - 10

Decision Tree

Q)

instance	a2	a3	classification
1	Hot	High	N
2	Hot	Wild	N
6	Cool	H	N
7	Hot	H	N
8	Hot	Normal	Yes

Entropy (S) = $-p_1 \log_2 p_1 - p_2 \log_2 p_2$

$= -\frac{1}{5}\log_2\left(\frac{1}{5}\right) - \frac{4}{5}\log_2\left(\frac{4}{5}\right)$

$= (0.2)(-2.32) - (0.8)(-0.32)$

$= 0.6646 + 0.256$

$S = 0.72$

a2 : Hot → 4 (3No 1Yes)

Cool → 1 (1 No)

$P(No) = \frac{3}{6}$

$P(Yes) = 1/6$

Entropy(Hot) = $-3/6 \log_2 3/6 - 1/6 \log_2 1/6$

$= 0.311 + 0.5$

$= 0.811$

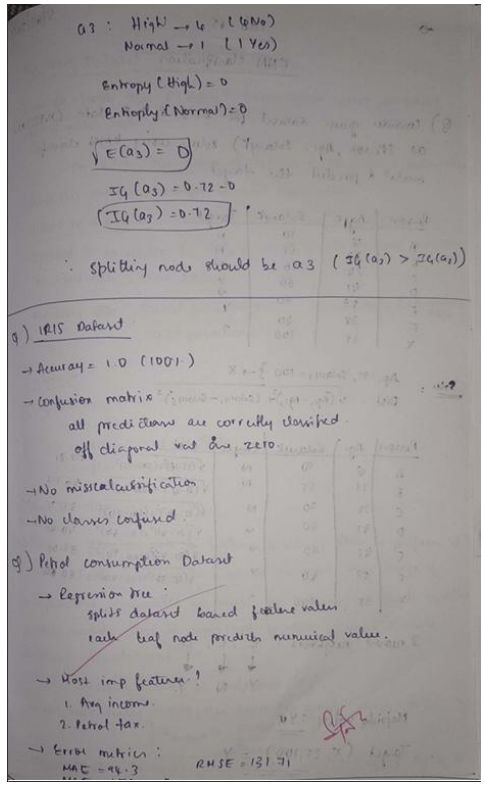
Entropy (Cool) = 0 (all are No)

$E(a2) = \frac{4}{6}(0.811) + \frac{1}{4}(0)$

$= 0.6488$

Info Gain (a2) = $0.72 - 0.6488$

$= 0.071$



a3 : High → 4 (4No)

Normal → 1 (1 Yes)

Entropy (High) = 0

Entropy (Normal) = 0

$E(a_3) = 0$

$IG(a_3) = 0.72 - 0$

$IG(a_3) = 0.72$

∴ Splitting node should be a3 ($IG(a_3) > IG(a_2)$)

Q) IRIS Dataset

→ Accuracy = 1.0 (100%)

→ Confusion matrix

all predictions are correctly classified

off diagonal vat are zero

→ No misclassification

→ No classes confused

Q) Petrol consumption Dataset

→ Regression tree

splits dataset based feature values

each leaf node predicts numerical value.

→ Most imp features :

1. Avg income

2. Petrol tax.

→ Error matrix :

MAE = 96.3 RMSE = 131.71

Code:

 # Decision Tree - IRIS Dataset

```
# Import Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# - - - - - - - - - - - - - - - - - - -
# Load Dataset
# - - - - - - - - - - - - - - - - - - -
iris = pd.read_csv("iris.csv")

print("First 5 rows:")
print(iris.head())
```

```
# ------------------
# Features and Target
# ------------------
X = iris.drop("species", axis=1)  # change column name if different
y = iris["species"]

# ------------------
# Train-Test Split (80-20)
# ------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# ------------------
# Build Model
# ------------------
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# ------------------
# Prediction
# ------------------
y_pred = model.predict(X_test)

# ------------------
# Accuracy
# ------------------
accuracy = accuracy_score(y_test, y_pred)
print("\nIRIS Dataset Accuracy:", accuracy)

# ------------------
# Confusion Matrix
# ------------------
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix - IRIS Dataset")
plt.show()
```

 # Decision Tree - Drug Dataset

```
# Import Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# ------------------
# Load Dataset
# ------------------
drug = pd.read_csv("drug.csv")

print("First 5 rows:")
print(drug.head())

# ------------------
# Data Preprocessing
# ------------------

# Encode categorical columns
le = LabelEncoder()

for column in drug.columns:
    if drug[column].dtype == 'object':
        drug[column] = le.fit_transform(drug[column])

# ------------------
# Features and Target
# ------------------
X = drug.drop("Drug", axis=1)  # change if column name different
y = drug["Drug"]

# ------------------
# Train-Test Split (80-20)
# ------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# ------------------
# Build Model
# ------------------
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# ------------------
# Prediction
# ------------------
y_pred = model.predict(X_test)

# ------------------
# Accuracy
# ------------------
```

```
accuracy = accuracy_score(y_test, y_pred)
print("\nDrug Dataset Accuracy:", accuracy)

# ------------------
# Confusion Matrix
# ------------------
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix - Drug Dataset")
plt.show()
```

 # Regression Tree - Petrol Consumption Dataset

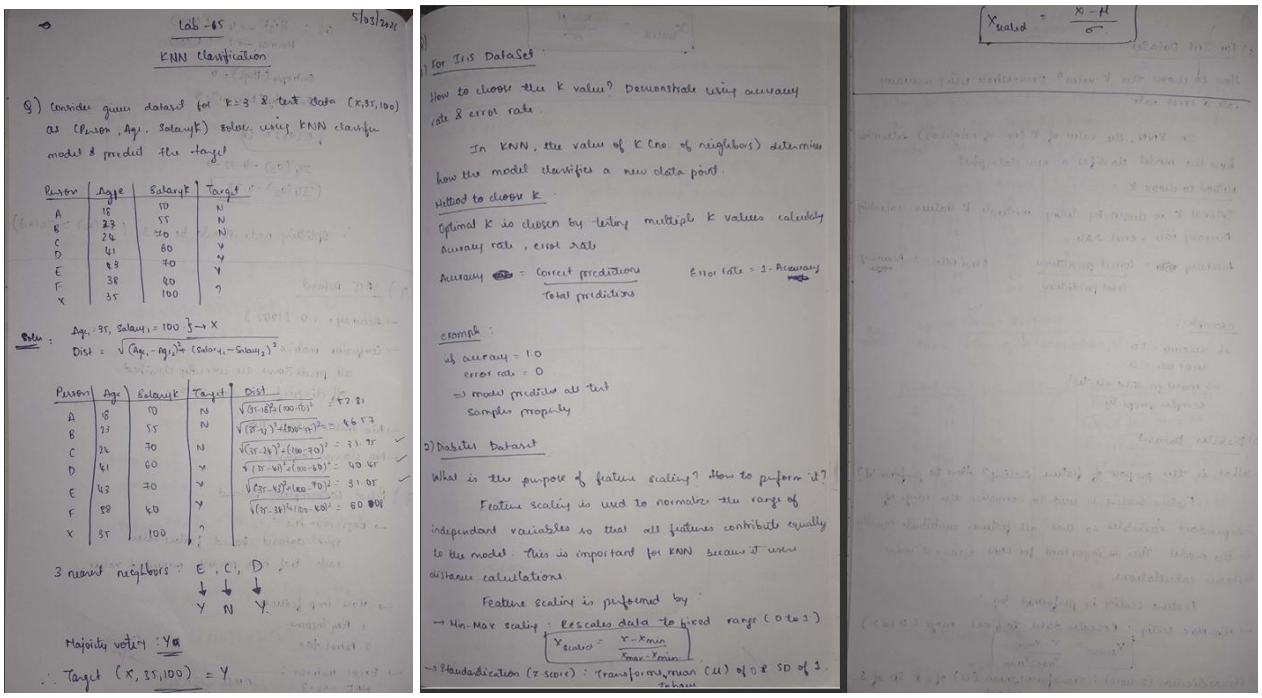
```
# Import Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error

# -----------------------
# Load Dataset
# -----------------------
data = pd.read_csv("petrol_consumption.csv")

print("First 5 rows:")
print(data.head())

# -----------------------
# Features and Target
# -----------------------
# Assuming last column is Petrol_Consumption
X = data.iloc[:, :-1]   # All columns except last
y = data.iloc[:, -1]    # Last column (target)

# -----------------------
# Train-Test Split (80% - 20%)
# -----------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# -----------------------
# Build Regression Tree Model
# -----------------------
regressor = DecisionTreeRegressor(random_state=42)
regressor.fit(X_train, y_train)
```

```
# ------------------------
# Prediction
# ------------------------
y_pred = regressor.predict(X_test)

# ------------------------
# Evaluation Metrics
# ------------------------
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)

print("\nModel Evaluation Metrics:")
print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
```

Program 6

Build KNN Classification model for a given dataset.

Screenshot



Code:

```
# KNN Classifier for IRIS Dataset
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
iris = pd.read_csv("iris.csv")

# Features and target
X = iris.drop("species", axis=1)  # change if target column name differs
y = iris["species"]

# Train-Test Split (80-20)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# KNN Model
k = 5
```

```
model = KNeighborsClassifier(n_neighbors=k)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

# Classification Report
print("Classification Report:\n", classification_report(y_test, y_pred))

# Plot Confusion Matrix
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix - Iris Dataset")
plt.show()


# KNN Classifier for Diabetes Dataset (With Feature Scaling)
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
diabetes = pd.read_csv("diabetes.csv")

# Features and target
X = diabetes.drop("Outcome", axis=1)
y = diabetes["Outcome"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# KNN Model
k = 7
model = KNeighborsClassifier(n_neighbors=k)
```

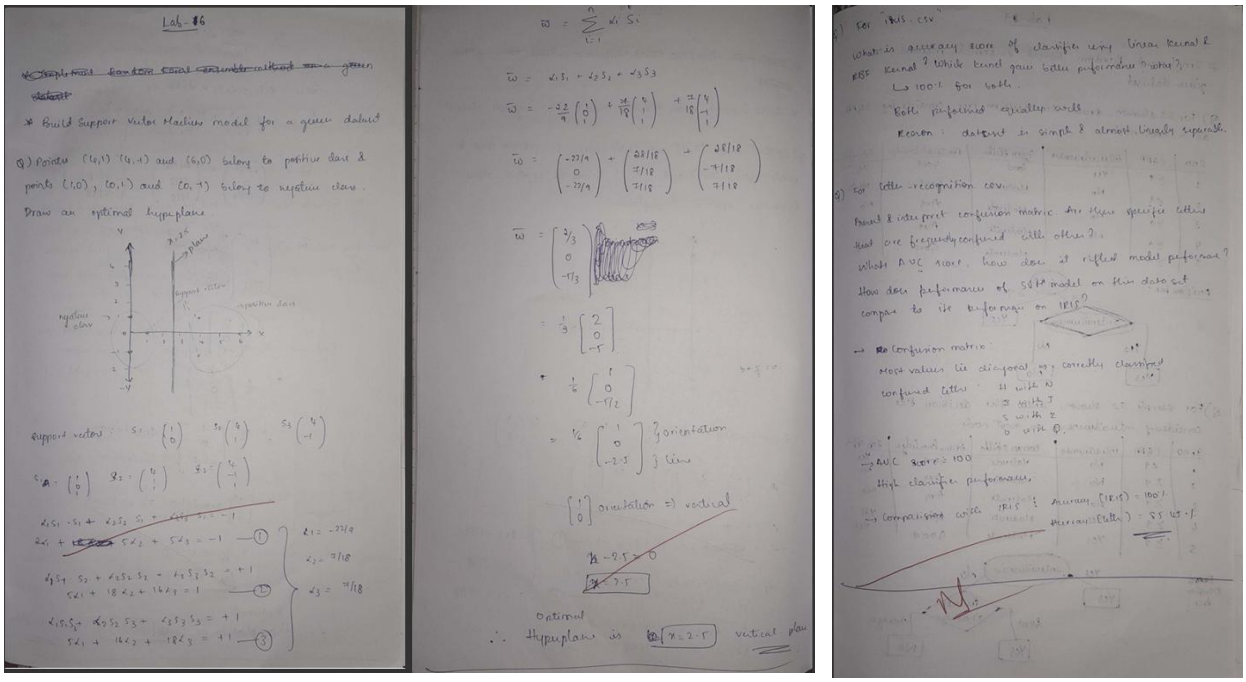
```
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)


# KNN Classifier for Heart Dataset (Finding Best K)
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load dataset
heart = pd.read_csv("heart.csv")

# Features and target
X = heart.drop("target", axis=1)
y = heart["target"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Find best K
scores = []

for k in range(1,21):
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)
    pred = model.predict(X_test)
    scores.append(accuracy_score(y_test, pred))

# Plot K vs Accuracy
```

```
plt.plot(range(1,21), scores)
plt.xlabel("K value")
plt.ylabel("Accuracy")
plt.title("Finding Best K")
plt.show()

# Best K
best_k = scores.index(max(scores)) + 1
print("Best K:", best_k)

# Train final model
model = KNeighborsClassifier(n_neighbors=best_k)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

# Plot Confusion Matrix
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix - Heart Dataset")
plt.show()

# Classification Report
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Program 7

Build Support vector machine model for a given dataset

Screenshot



Code:

```
# SVM Classifier for IRIS Dataset (Linear & RBF Kernel)
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix

# Load IRIS dataset
iris = pd.read_csv("iris.csv")

# Display dataset
print(iris.head())

# Features and target
X = iris.drop("species", axis=1)
y = iris["species"]

# Train-test split (80% training, 20% testing)
```

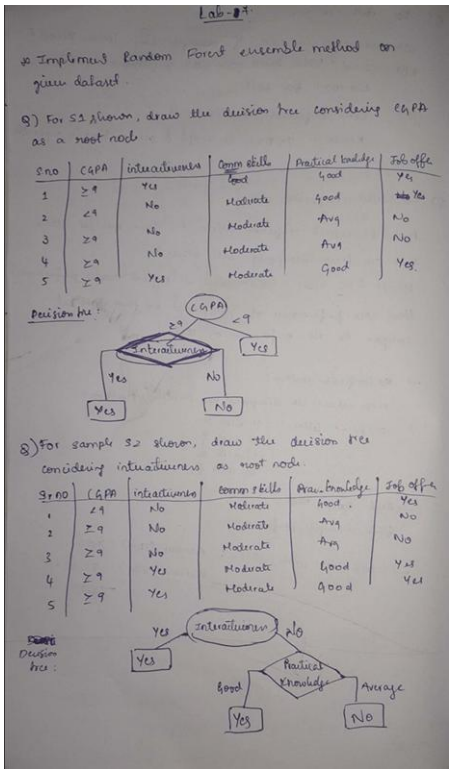
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# - - - - - - - - - - - - - - - - - - -
# SVM using RBF kernel
# - - - - - - - - - - - - - - - - - - -
rbf_model = SVC(kernel='rbf')

rbf_model.fit(X_train, y_train)

y_pred_rbf = rbf_model.predict(X_test)

print("RBF Kernel Accuracy:", accuracy_score(y_test, y_pred_rbf))

print("Confusion Matrix (RBF):")
print(confusion_matrix(y_test, y_pred_rbf))


# - - - - - - - - - - - - - - - - - - -
# SVM using Linear kernel
# - - - - - - - - - - - - - - - - - - -
linear_model = SVC(kernel='linear')

linear_model.fit(X_train, y_train)

y_pred_linear = linear_model.predict(X_test)

print("Linear Kernel Accuracy:", accuracy_score(y_test, y_pred_linear))

print("Confusion Matrix (Linear):")
print(confusion_matrix(y_test, y_pred_linear))
# SVM Classifier for Letter Recognition Dataset + ROC + AUC
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.preprocessing import label_binarize
from sklearn.metrics import roc_curve, auc

# Load dataset
letter = pd.read_csv("letter-recognition.csv")

print(letter.head())
```

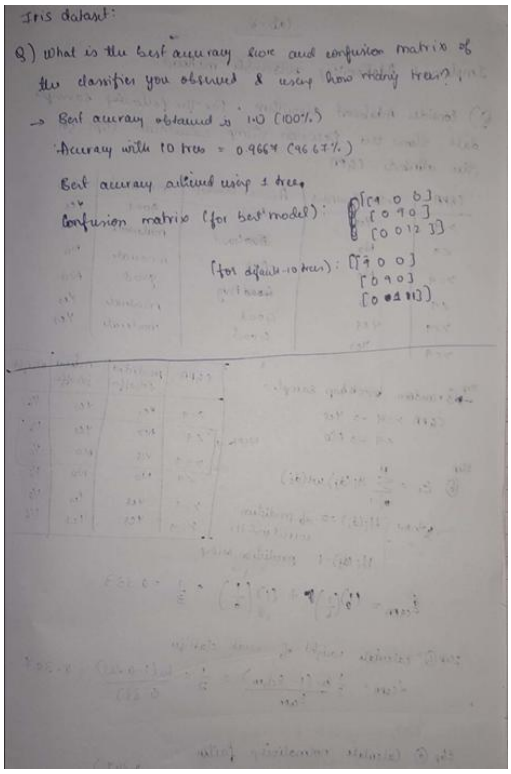
```
# Features and target
X = letter.iloc[:,1:]   # features
y = letter.iloc[:,0]    # letter label

# Train-test split (80% training, 20% testing)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Train SVM model
model = SVC(kernel='linear', probability=True)
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))
# Confusion Matrix
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# - - - - - - - - - - - - - - - - - - -
# ROC Curve and AUC Score
# - - - - - - - - - - - - - - - - - - -

# Convert labels to binary for ROC
classes = sorted(y.unique())
y_test_bin = label_binarize(y_test, classes=classes)

# Predict probabilities
y_score = model.predict_proba(X_test)

# ROC for first class
fpr, tpr, _ = roc_curve(y_test_bin[:,0], y_score[:,0])
roc_auc = auc(fpr, tpr)

# Plot ROC Curve
plt.figure()
plt.plot(fpr, tpr, label="ROC curve (AUC = %0.2f)" % roc_auc)
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

print("AUC Score:", roc_auc)
```

Program 8

Implement Random forest ensemble method on a given dataset.

Screenshot





Code:

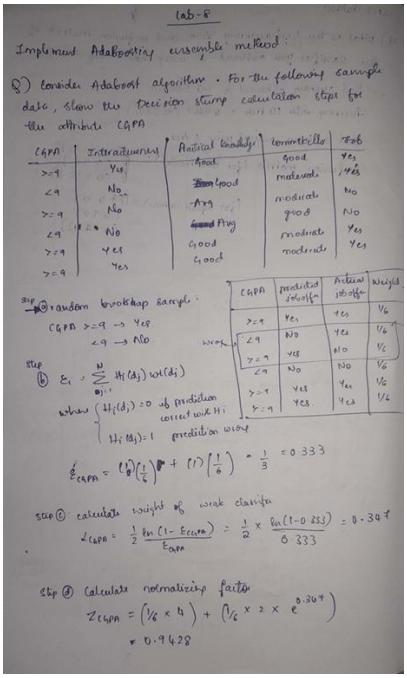
```
# Import required libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Load dataset
data = pd.read_csv("iris.csv")

# Features and target
X = data.drop("species", axis=1)  # assuming target column is named 'species'
y = data["species"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=68
)

# - - - - - - - - - - - - - - - - - - -
```

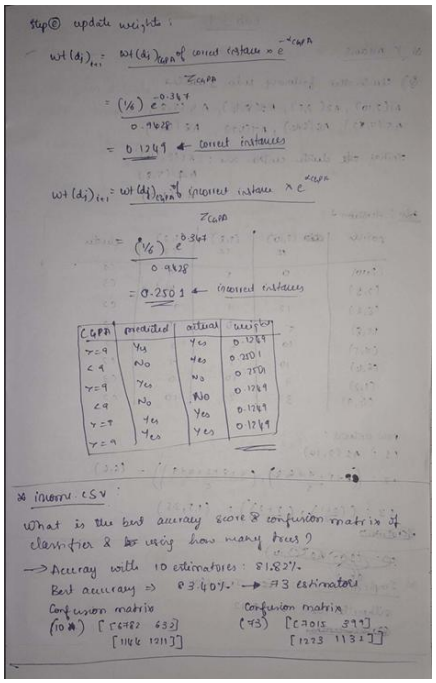
```
# 1. Default Random Forest (n_estimators = 10)
# - - - - - - - - - - - - - - - - - - - -
rf_default = RandomForestClassifier(n_estimators=10, random_state=42)
rf_default.fit(X_train, y_train)

y_pred_default = rf_default.predict(X_test)
default_score = accuracy_score(y_test, y_pred_default)

print("Accuracy with default (10 trees):", default_score)


# - - - - - - - - - - - - - - - - - - - -
# 2. Hyperparameter tuning (number of trees)
# - - - - - - - - - - - - - - - - - - - -
best_score = 0
best_n = 0

for n in range(1, 101):   # try 1 to 100 trees
    rf = RandomForestClassifier(n_estimators=n, random_state=42)
    rf.fit(X_train, y_train)

    y_pred = rf.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    if score > best_score:
        best_score = score
        best_n = n

print("Best accuracy:", best_score)
print("Best number of trees:", best_n)
from sklearn.metrics import confusion_matrix

# For default model
cm_default = confusion_matrix(y_test, y_pred_default)
print("Confusion Matrix (Default - 10 trees):\n", cm_default)

# For best model (inside loop OR after finding best_n)
rf_best = RandomForestClassifier(n_estimators=best_n, random_state=42)
rf_best.fit(X_train, y_train)

y_pred_best = rf_best.predict(X_test)
cm_best = confusion_matrix(y_test, y_pred_best)

print("\nConfusion Matrix (Best Model):\n", cm_best)
```

Program 9

Implement Boosting ensemble method on a given dataset.

Screenshot





Code:

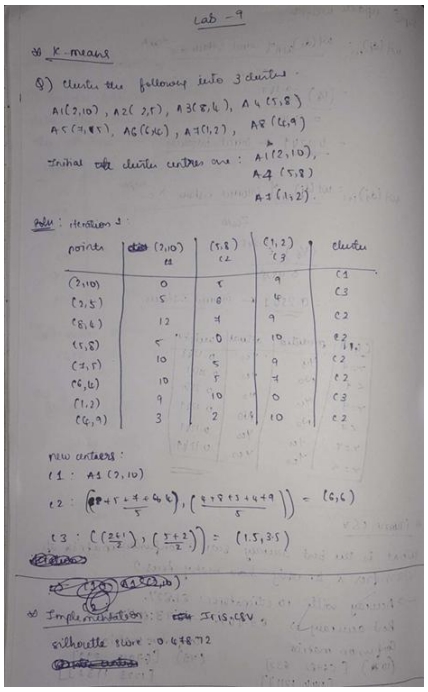
```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
data = pd.read_csv("income.csv")

# Features and target
X = data.drop("income_level", axis=1)
y = data["income_level"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ---------------------
# 1. AdaBoost with 10 estimators
# ---------------------
```

```
ada_default = AdaBoostClassifier(n_estimators=10, random_state=42)
ada_default.fit(X_train, y_train)

y_pred_default = ada_default.predict(X_test)
default_score = accuracy_score(y_test, y_pred_default)

print("Accuracy with 10 estimators:", default_score)

cm_default = confusion_matrix(y_test, y_pred_default)
print("Confusion Matrix (Default):\n", cm_default)


# - - - - - - - - - - - - - - - - - - - -
# 2. Hyperparameter tuning
# - - - - - - - - - - - - - - - - - - - -
best_score = 0
best_n = 0
best_cm = None

for n in range(1, 101):
    ada = AdaBoostClassifier(n_estimators=n, random_state=42)
    ada.fit(X_train, y_train)

    y_pred = ada.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    if score > best_score:
        best_score = score
        best_n = n
        best_cm = confusion_matrix(y_test, y_pred)

print("\nBest Accuracy:", best_score)
print("Best number of estimators:", best_n)
print("Confusion Matrix (Best Model):\n", best_cm)
```

Program 10

Build k-Means algorithm to cluster a set of data stored in a .CSV file.

Screenshot



Lab - 9

K-means

Q) Cluster the following into 3 clusters.

A1(2,10), A2(2,5), A3(8,4), A4(5,8)
A5(7,5), A6(6,4), A7(1,2), A8(4,9)

Initial cluster centers are: A1(2,10), A4(5,8), A7(1,2).

Soln: iteration 1:

points	(2,10) c1	(5,8) c2	(1,2) c3	cluster
(2,10)	0	5	9	c1
(2,5)	5	10	4	c3
(8,4)	12	7	9	c2
(5,8)	5	0	10	c2
(7,5)	10	5	9	c2
(6,4)	10	5	7	c2
(1,2)	9	10	0	c3
(4,9)	3	2	10	c2

new centers:

c1 : A1 (2,10)

c2 : $\left(\frac{8+5+7+6+4}{5}\right), \left(\frac{4+8+5+4+9}{5}\right)) = (6,6)$

c3 : $\left(\frac{2+1}{2}\right), \left(\frac{5+2}{2}\right)) = (1.5, 3.5)$

Implementation: Iris.csv.

silhouette score: 0.478772

Code:

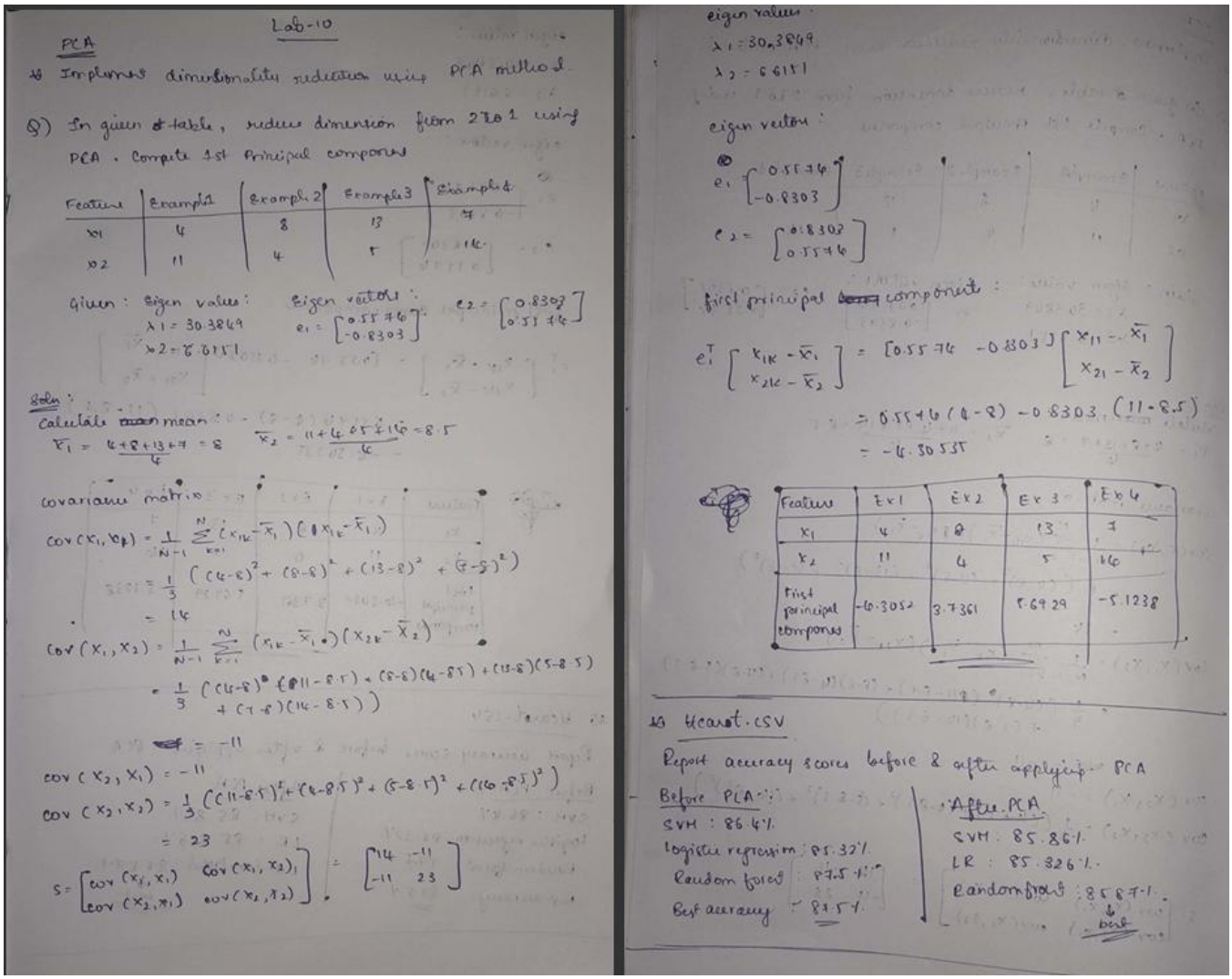
```
# Import libraries
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

# Load dataset
data = pd.read_csv("iris.csv")

# ----------------------------------
# Preprocessing
# ----------------------------------

# Drop target column if present (unsupervised learning)
if "species" in data.columns:
    X = data.drop("species", axis=1)
else:
    X = data

# Optional: Feature scaling (recommended for K-Means)
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)

# ----------------------------------
# 1. Apply K-Means (k = 3)
# ----------------------------------
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X_scaled)

labels = kmeans.labels_

print("Cluster labels:\n", labels)

# ----------------------------------
# 2. Evaluate clustering
# ----------------------------------
score = silhouette_score(X_scaled, labels)
print("Silhouette Score:", score)

# ----------------------------------
# 3. Cluster centers
# ----------------------------------
print("Cluster Centers:\n", kmeans.cluster_centers_)
```

Program 11

Implement Dimensionality reduction using Principal Component Analysis (PCA) method.

Screenshot



Lab-10

PCA

1) Implement dimensionality reduction using PCA method

Q) In given table, reduce dimension from 2 to 1 using PCA. Compute 1st principal component

Feature	Example 1	Example 2	Example 3	Example 4
x1	4	8	13	7
x2	11	4	5	14

Given: Eigen values: $\lambda_1 = 30.3849$, $\lambda_2 = 6.6151$

Eigen vectors: $e_1 = \begin{bmatrix} 0.5576 \\ -0.8303 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0.8303 \\ 0.5576 \end{bmatrix}$

Soln:

Calculate mean: $\bar{x}_1 = \frac{4+8+13+7}{4} = 8$, $\bar{x}_2 = \frac{11+4+5+14}{4} = 8.5$

covariance matrix

$$cov(x_1, x_1) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{1k} - \bar{x}_1)(x_{1k} - \bar{x}_1)$$

$$= \frac{1}{3} \left((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2 \right)$$

$$= 14$$

$$cov(x_1, x_2) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{1k} - \bar{x}_1)(x_{2k} - \bar{x}_2)$$

$$= \frac{1}{3} \left((4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5) \right)$$

$$= -11$$

$$cov(x_2, x_1) = -11$$

$$cov(x_2, x_2) = \frac{1}{3} \left((11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2 \right)$$

$$= 23$$

$$S = \begin{bmatrix} cov(x_1, x_1) & cov(x_1, x_2) \\ cov(x_2, x_1) & cov(x_2, x_2) \end{bmatrix} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

eigen values:

$\lambda_1 = 30.3849$

$\lambda_2 = 6.6151$

eigen vectors:

$e_1 = \begin{bmatrix} 0.5576 \\ -0.8303 \end{bmatrix}$

$e_2 = \begin{bmatrix} 0.8303 \\ 0.5576 \end{bmatrix}$

first principal component:

$$e_1^T \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 0.5576 & -0.8303 \end{bmatrix} \begin{bmatrix} x_{11} - \bar{x}_1 \\ x_{21} - \bar{x}_2 \end{bmatrix}$$

$$= 0.5576(4-8) - 0.8303(11-8.5)$$

$$= -4.3055$$

Feature	Ex 1	Ex 2	Ex 3	Ex 4
x_1	4	8	13	7
x_2	11	4	5	14
First principal component	-4.3055	3.7361	5.6729	-5.1238

2) Heart.csv

Report accuracy scores before & after applying PCA

Before PCA:
SVM : 86.4%
Logistic regression : 85.32%
Random forest : 87.5%
Best accuracy : 87.5%

After PCA
SVM : 85.86%
LR : 85.32%
Random forest : 85.87% ↓ best

Code:

```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA

# Models
from sklearn.svm import SVC
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# Load dataset
data = pd.read_csv("heart.csv")

# ---------------------
# Features and target
# ---------------------
X = data.drop("HeartDisease", axis=1)
y = data["HeartDisease"]

# ---------------------
# Column separation
# ---------------------
categorical_cols = ["Sex", "ChestPainType", "RestingECG", "ExerciseAngina", "ST_Slope"]
numeric_cols = ["Age", "RestingBP", "Cholesterol", "FastingBS", "MaxHR", "Oldpeak"]

# ---------------------
# Preprocessing
# ---------------------
preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numeric_cols),
        ("cat", OneHotEncoder(), categorical_cols)
    ]
)

# ---------------------
# Train-test split
# ---------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ---------------------
# Models
# ---------------------
models = {
    "SVM": SVC(),
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Random Forest": RandomForestClassifier()
}

# ---------------------
# BEFORE PCA
# ---------------------
print("---- BEFORE PCA ----")
```

```
best_score = 0

for name, model in models.items():
    pipe = Pipeline([
        ("preprocess", preprocessor),
        ("classifier", model)
    ])

    pipe.fit(X_train, y_train)
    y_pred = pipe.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    print(f"{name}: {score}")

    if score > best_score:
        best_score = score

print("Best Accuracy before PCA:", best_score)

# ---------------------
# AFTER PCA
# ---------------------
print("\n---- AFTER PCA ---- ")

pca = PCA(n_components=0.95)

best_score_pca = 0

for name, model in models.items():
    pipe = Pipeline([
        ("preprocess", preprocessor),
        ("pca", pca),
        ("classifier", model)
    ])

    pipe.fit(X_train, y_train)
    y_pred = pipe.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    print(f"{name}: {score}")

    if score > best_score_pca:
        best_score_pca = score

print("Best Accuracy after PCA:", best_score_pca)
```